

Indian Institute of Technology Jodhpur
Department of Physics
End-Sem Exam (PHL 1010)

Duration: 3 Hours

24/04/2026

Max. Marks: 60

Instructions:

- This paper carries seven questions. Attempt all questions. You may like to use formula given at the end of the paper.
- Keep solution of all parts together for the same question. Start every question on a fresh paper.

1. Complete the following:

- (a) Total energy streaming through a space in form of EM wave is _____
- (b) Total number of image charges for a point charge kept between two semi-infinite conducting planes, inclined at an angle θ will be _____
- (c) Electric field inside a uniformly polarized sphere is $-\vec{P}/3\epsilon_0$. Hence the magnetic field inside a uniformly magnetized sphere would be _____
- (d) $\vec{H} = (2\hat{x} - \hat{y} + 5\hat{z})\cos 333t$ in a region of relative permeability 100. The interface between this region and free space is $z = 0$ plane and there is no current on the interface. Magnetic field $\vec{B}$ in free space is _____ [1+1+1+3]

2. (a) Electric field associated with an EM wave is given by $\vec{E} = 10^{-3}(\hat{z} - \hat{y})\sin[10^7(x + y + z - \beta t)] V/m$. Determine the direction of propagation of the wave. Is the wave transverse? Find β and angular frequency ω .

(b) Electric fields associated with two EM waves propagating in a medium are given by

$$\vec{E}_1 = \hat{x} E_0 \cos(\omega t - kz + 5\pi/12)$$

$$\vec{E}_2 = \hat{y} E_0 \cos(\omega t - kz - \pi/12)$$

(i) Find out the nature of resulting field. (ii) Find out the corresponding magnetic field and energy carried per unit time, per unit area by the field. [2+6]

3. (a) Suppose the field inside a large piece of magnetic material is $\vec{B}_0$, so that $\vec{H}_0 = (1/\mu_0)\vec{B}_0 - \vec{M}$, where $\vec{M}$ is a "frozen-in" magnetization. Now a long needle-shaped cavity running parallel to $\vec{M}$ is hollowed out of the material (Fig. 1). Find the field at the center of the cavity, in terms of $\vec{B}_0$ and $\vec{M}$. Also find $\vec{H}$ at the center of the cavity, in terms of $\vec{H}_0$ and $\vec{M}$. [4]

(b) A long solid metal cylinder rotates with angular velocity about its axis of symmetry. The cylinder is in a homogeneous magnetic field $\vec{B}$ parallel to its axis. What is the resultant charge distribution inside the cylinder? Is there an angular velocity for which charge density is everywhere zero? [6]

Figure 1

4. A conducting sphere (radius a), centered at the origin, is placed in a uniform electric field $\vec{E} = E_0 \hat{y}$. Uniform field near the origin and in the region of sphere (whose radius is very small) can be thought of as being produced by two equal and opposite charges $+q$ and $-q$ located at $y = -D$ and $+D$ from the centre of the sphere (D being very large). Using the method of images, find the potential at a general point outside the conductor. Use this result to find induced charge density. Plot the charge density and find total induced charge on the conducting sphere. [6]

5. (a) A conducting (homogeneous) spherical shell of radius R carrying a charge $+Q$, is attached to one end of an elastic string. A $+q$ charge is placed at the other end of the string and the sphere is hanging in the gravitational field of the earth.

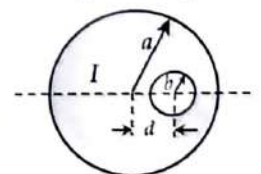
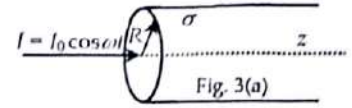


Figure 2

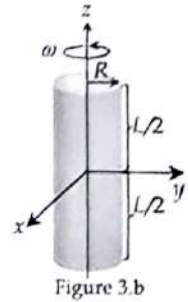
Write down the equation of motion that can be used to calculate the extension of the string.
[3]

- (b) A long straight cylindrical conductor of radius a has a hole of radius b along its length. Axis of hole is shifted by a distance d from the axis of the conductor (Figure 2). The conductor carries a uniform current I which flows out of the paper. Calculate magnetic field inside the hole. [5]

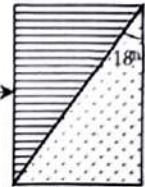
6. (a) A cylindrical conductor of radius R and infinite length carries a current $I = I_0 \cos \omega t$, which is uniformly distributed throughout the cross section (Fig. 3.a). The conductivity of the conductor is σ . Find the time-averaged dissipated power per unit length in the conductor (neglect displacement currents in the conductor, if any).



- (b) A thin glass rod of radius R and length L carries a uniform surface charge σ . It is set spinning about its axis at an angular velocity ω . Find the magnetic field at a distance $r \gg R$ from the center of the rod (Fig. 3.b). [You may like to consider it as current carrying loops, each of radius R kept over each other occupying length L . Also, you may consider the point point on the x-axis] [3+7]



7. (a) A Wollaston prism made of calcite is shown in Fig. 4a. If an unpolarized light falls on this prism at a normal incidence, draw the ray propagation through the prism depicting the field vibrations. Determine the angle of deviation between the two emergent beams if the prism angle is 18° and $n_o = 1.658$, $n_e = 1.486$.



- (b) An unpolarized beam falls at a linear polarizer (LP) whose pass-axis (PA) makes 60° from the vertical. Light emerging from LP falls on a Quarter Wave Plate (QWP) made of Quartz whose optical axis is horizontal (Fig. 4b). Light emerging from QWP passes through another LP with PA vertical. Write down the electric field after the LP, after the QWP and then again from LP. Determine the output state-of-polarization after each element. [5 + 7]

